THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 6: **Triangles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In $\triangle ABC$, $AB = 7$, $BC = 8$, $AC = 10$. The triangle is: 1

(a) isosceles right **(b)** equilateral **(c)** right-angled **(d)** not right-angled

2. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 3 : 5$. Then $AE : AC$ is 3: 1

(a) 5 **(b)** 15 **(c)** 9 **(d)** 8

3. In $\triangle ABC$, $DE \parallel BC$, $AD = 5$ cm, $DB = 4$ cm, $AE = 15$ cm. Then EC (in cm) $=$ 1

(a) 20 **(b)** 12 **(c)** 15 **(d)** 4

4. $\triangle ABC \sim \triangle DEF$ with areas 9 cm² and 36 cm². If $AB = 3$ cm, then DE (cm) $=$ 1

(a) 3 **(b)** 12 **(c)** 6 **(d)** 9

5. The length of the diagonal of a square of side 6 cm, squared, is: 1

(a) 144 **(b)** 36 **(c)** 12 **(d)** 72

6. A man goes 9 m due east and then 12 m due north. His distance from the start (m) is: 1

(a) 15 **(b)** 225 **(c)** 21 **(d)** 3

7. In $\triangle ABC$, $AB = 8$, $BC = 15$, $AC = 17$. The triangle is: 1

(a) isosceles right **(b)** equilateral **(c)** not right-angled **(d)** right-angled

8. In $\triangle ABC$, $DE \parallel BC$ and $AD : DB = 1 : 4$. Then $AE : AC$ is 1: 1

(a) 1 **(b)** 6 **(c)** 4 **(d)** 5

9. In $\triangle ABC$, $DE \parallel BC$, $AD = 2$ cm, $DB = 9$ cm, $AE = 8$ cm. Then EC (in cm) $=$ 1

(a) 36 **(b)** 18 **(c)** 8 **(d)** 9

10. $\triangle ABC \sim \triangle DEF$ with areas 12 cm² and 75 cm². If $AB = 6$ cm, then DE (cm) $=$ 1

(a) 15 **(b)** 18 **(c)** 6 **(d)** 75

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Sides of a triangle are 6 cm, 7 cm, 10 cm. State with reason whether it is right-angled. 2

12. The perimeters of two similar triangles are 24 cm and 72 cm. Find the ratio of their corresponding sides and their areas. 2

13. A ladder 29 m long reaches a window 21 m high. Find how far the foot of the ladder is from the wall. 2

14. Sides of a triangle are 13 cm, 84 cm, 85 cm. State with reason whether it is right-angled. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AD = 2$ cm, $DB = 7$ cm and $AE = 10$ cm, find EC. 3

16. The areas of two similar triangles are 25 cm² and 64 cm². If a side of the first is 5 cm, find the corresponding side of the second. 3

17. In $\triangle ABC$, $DE \parallel BC$ with $AD : DB = 1 : 3$. Find the ratio of the area of $\triangle ADE$ to that of $\triangle ABC$, and to that of trapezium $DBCE$. 3

OR

In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AD = 6$ cm, $DB = 4$ cm and $AE = 12$ cm, find EC. 3

18. The areas of two similar triangles are 24 cm² and $\frac{200}{3}$ cm². If a side of the first is 6 cm, find the corresponding side of the second. 3

OR

In $\triangle ABC$, $DE \parallel BC$ with $AD : DB = 3 : 4$. Find the ratio of the area of $\triangle ADE$ to that of $\triangle ABC$, and to that of trapezium $DBCE$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that the ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides. 5

OR

Two similar triangles have perimeters 15 cm and 60 cm. If a side of the smaller is 4 cm, find the corresponding side of the larger and the ratio of their areas. 5

20. Prove that in a right triangle, the square of the hypotenuse equals the sum of the squares of the other two sides. 5

OR

Two similar triangles have perimeters 12 cm and 60 cm. If a side of the smaller is 3 cm, find the corresponding side of the larger and the ratio of their areas. 5

END OF QUESTION PAPER — Total: 40 Marks

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